PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · MATHEMATICS B · 4MB1/02

A* Examination Guide

Built from 16 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the Mathematics B Paper 2 question papers and mark schemes. Methods are laid out the way the mark schemes award M, A and B marks, and every warning comes from a mark-scheme note such as “working required”, “do not ISW”, “SC” or “correct answer from obvious incorrect working scores 0”. The formulae sheet is printed in the paper; everything else must be learnt.

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Papers analysed: Jan 2019 (legacy 4MB0), Jun 2019, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 HOW PAPER 2 IS BUILT

Format: 2 hours 30 minutes, 100 marks, 11–12 long structured questions of 7–14 marks each, calculator allowed: 1.5 minutes per mark. There is no formulae sheet, but the formula a question needs (sine rule, cosine rule, matrix inverse, quadratic formula, cone volume) is often printed in a box beside it. Every paper has a **transformations-with-matrices** question and a **money/percentage context** question; most have a Venn diagram, a histogram, a vector question, non-linear simultaneous equations and a graph or kinematics question.

Topic	Papers (of 16)	What is asked most
Transformations and matrices on a grid	16	draw images, describe fully, find the matrix N, area factor
Money, percentages, ratio, exchange rates	16	profit, reverse %, compound interest, currency comparison
Statistics (grouped mean, histogram)	15	mean from a table, median class, complete a histogram, probability
Probability (trees, without replacement)	14	complete a tree, unknown n from a given probability, conditional
Sets and Venn diagrams	14	3-set Venn with x, list members, n(A ∩ B′), conditional
Vectors	13	ratio along a line, collinearity proof, area ratio (6–8 marks)
Circles, angles and 3D trigonometry	13	angle reasons, tangents, sectors, angle between line and plane
Non-linear simultaneous equations	9	line with circle or quadratic; show a line is a tangent
Graphs (plot, draw a line, regions)	10	complete table, smooth curve, solve by drawing a line; shade R

Topic	Papers (of 16)	What is asked most
Calculus and kinematics	9	v and a from x(t), at rest, distance, gradient conditions, minimum S
Bounds	8	upper/lower bound of area, speed, height; "definitely enough"
Mensuration and optimisation	8	prisms, pyramids, cone + hemisphere, minimum surface area

2 TWENTY MARK-SCHEME RULES

1. **"Show clear algebraic working":** correct answers with no working score no marks (simultaneous equations, quadratics, vector proofs).
2. **Describe a transformation fully with exactly one transformation:** rotation + angle + direction + centre; enlargement + scale factor + centre; reflection + line; translation + vector. Naming two transformations scores 0.
3. **Enlargement wording:** "scale factor" (or "SF") must appear; a negative factor needs the centre too (SF −2, centre (6, 2)).
4. **Matrix multiplication order:** "MN" means apply N first, then M. Set up the matrix product in the correct order.
5. **Find N from images:** N × (coordinates of D) = (coordinates of B), so $N = B\,D^{-1}$; show the matrices.
6. **Label every image** (B, C, D); missing labels are penalised once per question.
7. **Show a line is a tangent:** substitute, form the quadratic, show $b^2 - 4ac = 0$ and conclude "only one point of intersection".
8. **Vector proofs:** state the scalar multiple and the shared point for collinearity; for parallel lines show one vector is a multiple of the other.
9. **Angle reasons:** each calculated angle with its reason ("angle at the centre is twice the angle at the circumference", "alternate segment theorem", "tangent ⊥ radius").
10. **Graphs:** plot accurately (±½ square), join with a smooth curve, draw the straight line you use and read to 1 d.p.
11. **Regions:** draw each boundary line, shade or label R exactly as instructed; integer points must satisfy strict inequalities when asked.
12. **Kinematics:** differentiate x(t) for v, v(t) for a; "at rest" means v = 0; reject negative t.
13. **Bounds:** "definitely enough" requires the upper bound of what is needed vs the lower bound of what is available.
14. **Money:** give answers to 2 d.p.; compare costs in the same currency and state which is cheaper.
15. **Percentage of a total:** use the original as the base; reverse percentages divide by the multiplier.
16. **Histograms:** frequency density = frequency ÷ class width; label the axis; probabilities from area.
17. **Venn diagrams with algebra:** fill from the centre, write each region in x, then use the total.
18. **Probability equations:** form the expression in n, simplify to a quadratic, solve, and reject non-integer roots.
19. **Optimisation:** show $dS/dx = 0$, solve, and justify the minimum with $d^2S/dx^2 > 0$.
20. **Accuracy:** 3 s.f. unless told; angles to 1 d.p. when asked; keep full calculator values.

3 MONEY, PERCENTAGES, RATIO AND BOUNDS

- Ratio shares: 378 drinks in 5 : 3 : 1 ⇒ 42 per part; coffee − hot chocolate = 4 × 42 = 168 (Nov 20).
- Profit share (Jun 21): 20% + 80% × 55/80 = 75% of the profit.
- Reverse percentage: price after a 6.25% rise is 5.10 ⇒ 5.10 ÷ 1.0625 = 4.80 (Nov 21). Selling price £15.95 at 16% profit ⇒ cost 15.95 ÷ 1.16 = £13.75 (Jun 25).
- Compound interest (Nov 25): £20 000 × 1.084^2 = £23 501.12. Percentage profit = profit ÷ cost × 100.
- Tiered profit (Jun 23): compute each tier's profit, add, then divide by total selling price.
- Currency comparisons (Nov 20, Jan 23, Jun 26): convert both prices into one currency, include tax and postage, and state which is cheaper.
- Population figures in standard form (Jun 19, Jun 21): 48% of 2.35×10^7, then ratio 5 : 3 : 2.

BOUNDS

Upper bound of an area or volume uses all upper bounds; a quotient max = $\text{numerator}_{max} \div \text{denominator}_{min}$.

"Definitely enough paint" (Jun 22): compare the upper bound of the surface area with the paint coverage.

Longest rod that definitely fits (Jun 22): use lower bounds in the space diagonal $\sqrt{(l^2 + w^2 + h^2)}$.

Lower bound of a height from mass and density (Nov 23): $h = m_{min} \div (\rho_{max} \times \text{base}_{max})$.

4 ALGEBRA AND EQUATIONS

NON-LINEAR SIMULTANEOUS EQUATIONS (6 MARKS)

Nov 20: $x^2 + y^2 = 26$, $2x + y = 9 \Rightarrow y = 9 - 2x \Rightarrow 5x^2 - 36x + 55 = 0 \Rightarrow (5x - 11)(x - 5) = 0 \Rightarrow$ (5, −1) and (2.2, 4.6).

Tangent proof (Jan 22): $x^2 + y^2 = 64$ and $4y + 3x = 40 \Rightarrow 25x^2 - 240x + 576 = 0$, discriminant $240^2 - 4(25)(576) = 0$, so one point only: (4.8, 6.4).

- Index equations (Jun 22): $2^x \times 4^y = 128 \Rightarrow x + 2y = 7$; $8^x/4^y = 32 \Rightarrow 3x - 2y = 5 \Rightarrow x = 3, y = 2$.
- Hidden quadratics (Nov 23): $x^{2/3} - 3x^{1/3} - 10 = 0$ with $y = x^{1/3} \Rightarrow (y - 5)(y + 2) = 0 \Rightarrow x = 125$ or −8.
- Algebraic fractions (Nov 20): factorise $(25x^2 - 36 = (5x - 6)(5x + 6))$, invert the divisor, cancel.
- Subject of a formula (Jan 21): $c = 5(f - 32)/9 \Rightarrow f = 9c/5 + 32$.
- Simultaneous equations from a shape (Nov 23 kite): set equal sides equal, solve, then substitute for the perimeter.
- Proportion chains (Jan 21): $a = k/b$ and $b = m/c^3 \Rightarrow a = (k/m)c^3$; complete the table from one known pair.
- Functions (Jan 19): use given points to form equations in p and q; $ff(x) = p(px + q) + q$.

5 INEQUALITIES, REGIONS AND GRAPHS

- Linear and quadratic together (Jan 21): $3x + 5 > 7x - 9 \Rightarrow x < 3.5$; $x^2 + 5x - 6 \le 0 \Rightarrow -6 \le x \le 1$; both $\Rightarrow -6 \le x \le 1$.
- Regions (Nov 20, Nov 21, Jun 23, Nov 23): draw each line (solid for ≤/≥, dashed for), shade or label R; list integer points that satisfy all inequalities (and any extra line given).
- Plotting (Jun 21, Nov 21, Jun 23): complete the table to the stated accuracy, plot every point, smooth curve through them.
- Solving with a drawn line: rearrange so one side is the plotted function. With $y = x^2 + 2x + 4/x$ plotted, $x^2 + 2x + 4/x = 5x + 3 \Rightarrow$ draw $y = 5x + 3$ and read the x-coordinates (Jun 21).
- Line through two intersection points (Jun 23): read both points, find the gradient and the equation.
- Gradient condition (Jun 21): substitute the point and dy/dx at that point to get two equations in a and b.

6 MATRICES

TOOLKIT

Product AB: rows of A × columns of B. det(a b / c d) = ad − bc; inverse = (1/det)(d −b / −c a).

Solving with matrices (Jan 19): write the simultaneous equations as $M(x\ y)^T = (p\ q)^T$, so $(x\ y)^T = M^{-1}(p\ q)^T$.

Unknown matrix (Jan 22): $BA = C \Rightarrow B = CA^{-1}$; use the given det B to find the unknown element.

Area scale factor of a transformation = |det M| (Nov 20).

7 TRANSFORMATIONS

Transformation	Matrix	Transformation	Matrix
reflection in x-axis	(1 0 / 0 −1)	rotation 90° anticlockwise, O	(0 −1 / 1 0)
reflection in y-axis	(−1 0 / 0 1)	rotation 90° clockwise, O	(0 1 / −1 0)
reflection in y = x	(0 1 / 1 0)	rotation 180°, O	(−1 0 / 0 −1)
reflection in y = −x	(0 −1 / −1 0)	enlargement SF k, O	(k 0 / 0 k)

METHOD

Draw an image: multiply the matrix by the 2 × 3 matrix of vertex coordinates, plot, join and label.

Describe fully: one transformation only. Rotation: angle, direction, centre. Enlargement: scale factor (can be negative or fractional), centre. Reflection: equation of the line. Translation: column vector.

Find the centre: join corresponding points (enlargement) or use perpendicular bisectors (rotation).

Find N with N × D = B: N = (B's coordinates)(D's coordinates)$^{-1}$ using two vertices, or combine known matrices (Jun 24: "using a combination of transformation matrices").

- Enlargement with negative scale factor (Jun 21, Jan 22, Jun 26): image is on the opposite side of the centre and inverted.
- Combined transformation MN: apply N first. Area of image = |det(MN)| × area of original.
- Reflection repeated returns the original (Jan 21: $M^2 = I$).

8 CALCULUS AND KINEMATICS

WORKED EXAMPLES

Jan 19: $s = 4t^3 - 18t^2 + 5t + 10 \Rightarrow v = 12t^2 - 36t + 5$, $a = 24t - 36 > 0 \Rightarrow t > 1.5$.

Jun 24: $x = kt^2 - t^3$, $v(2) = 24 \Rightarrow 4k - 12 = 24 \Rightarrow k = 9$; at rest when $18t - 3t^2 = 0 \Rightarrow t = 6 \Rightarrow OA = 108$ m.

Jun 25 (minimum surface): $S = 4x^2 + 3a/x$; $dS/dx = 8x - 3a/x^2 = 0 \Rightarrow 3a = 8x^3 \Rightarrow S = 12x^2 = 400 \Rightarrow x = 5.77$, $a = 513$.

- Range of acceleration for a speed condition (Jun 26): solve v > value as a quadratic inequality in t, then find a at the ends of the interval.
- Stationary points: $dy/dx = 0$; nature from d^2y/dx^2. Minimum points of curves on graph questions can be checked algebraically.

9 GEOMETRY, CIRCLES AND 3D TRIGONOMETRY

- Tangent–secant (Jun 19): $TA^2 = TC \times TD \Rightarrow 100 = 8(8 + 2r) \Rightarrow r = 2.25$ cm; then arc length = $(\theta/360) \times 2\pi r$.
- Angles in circles (Jun 21, Jan 22): cyclic quadrilateral opposite angles 180°; angle at centre twice angle at circumference; alternate segment; tangent ⊥ radius — each with the reason.
- Kite (Jan 21): AC = 10 cm, area = 48 cm^2, BE = 48/10 = 4.8 cm; similar triangles by two equal angles with reasons.
- Circle chord QT in surd form (Jan 21): use the angle in a semicircle (90°) and Pythagoras without a calculator.
- 3D (Nov 21, Jun 22, Jun 24): pyramid slant heights from Pythagoras; angle of elevation needs the right-angled triangle with the vertical height; use the printed cosine rule for angles in non-right triangles.
- Sine and cosine rules, area ½ab sin C (printed); for x and y in a quadrilateral (Nov 20) split it into triangles.

10 MENSURATION AND OPTIMISATION

- Prisms: volume = cross-section × length; surface = all faces. Perimeter conditions give missing lengths (Nov 20).
- Composite solid (Nov 24): hemisphere : total = 8 : 25 with total 1660 ⇒ hemisphere 531.2 = $(2/3)\pi r^3 \Rightarrow r = 6.33$; cone 1128.8 = $(1/3)\pi r^2 h \Rightarrow h = 26.9$ cm.
- Packing (Jan 23): number along each edge = floor(edge ÷ item length), multiplied.
- Sectors (Nov 24): arc = $\theta/360 \times 2\pi r$; shaded ring areas are differences of sectors; equate to a fraction of a sector to find a radius.
- Polygons (Jun 23): exterior angle = 360/n; area of a regular polygon = n triangles of ½ab sin C.

11 VECTORS

METHOD (6–8 MARKS)

Write each route in terms of a and b; for a point on a line use a scalar (λ, k, μ); express the same vector two ways; equate the a and b coefficients; solve.

Collinear (Jun 21): show AP = k × PB (scalar multiple) and that they share P, then give the ratio m : n.

Parallel (Jun 22): DC = AB × constant shows AB ∥ DC; area ratios from the ratio of parallel sides.

Jan 19: OA = 6a, OB = 4b, OC : OA = 2 : 3 and CD ∥ AB ⇒ OD = (2/3)OB, so DB = (4/3)b.

12 STATISTICS

- Grouped mean: Σ(midpoint × f) ÷ Σf; median class from the (n/2)th value.
- Histogram (Jan 19, Nov 21, Nov 25): frequency density = f ÷ width; complete missing rows from the bars and vice versa.
- Histogram scaling (Jun 22): if a 30-minute class is a 3 cm square, 1 cm width = 10 minutes and the height scale follows.
- Probability from a histogram (Jan 19): between 18 and 34 minutes — take the fraction of each partial bar.
- Combined means (Jan 22): 34 × 49 − 11 × 72 = 874 for 23 people ⇒ mean 38.0 years.
- Pie charts: angle = frequency ÷ total × 360°.

13 PROBABILITY

- Trees: multiply along, add between; independent events keep the same probabilities.
- Unknown n (Jan 23, Nov 24, Jun 24): write the probability with n, set equal to the given value, form a quadratic, solve and reject impossible values. Example: P(both white) = 10/21 from 11 black and w white $\Rightarrow w(w-1)/((w+11)(w+10)) = 10/21$.
- Explaining impossibility (Jan 23): 28/(28 + n) cannot equal 5/11 if it forces a non-integer n.
- Conditional (Jun 25): given all three counters the same colour, divide P(all red) by P(all same).
- Expected number = probability × trials; with different daily probabilities (Jan 21) sum the expectations.

14 SETS AND VENN DIAGRAMS

- Notation: $A \cap B$, $A \cup B$, A', , n(A), $\subset$. List members exactly, without repeats.
- Three-set Venn with x (Jan 21): put x in the centre, pairs minus x, then singles; total = 100 gives y in terms of x; the greatest x comes from every region being ≥ 0.
- Algebraic regions (Jun 21): add all expressions, set equal to the total, solve the quadratic, reject negatives.
- Conditional from Venn: $P(C \mid P) = n(C \cap P) \div n(P)$; "given a multiple of 3" restricts the denominator.
- Number sets (Jan 23, Nov 23): odd numbers, primes, factors of 30 — list carefully before placing.

15 FINAL A* CHECKLIST

1. Transformations: one transformation, all details, every image labelled; MN means N first.
2. Matrices: inverse with 1/det; $N = B D^{-1}$; area factor |det|.
3. Simultaneous equations: substitute, quadratic, factorise or formula, pair the values; tangent $\Rightarrow$ discriminant 0.
4. Vectors: scalar routes, equate coefficients, state the conclusion.
5. Money: same currency, 2 d.p., state the comparison.
6. Bounds: choose upper/lower correctly for "definitely".
7. Graphs: smooth curve, drawn line, values to 1 d.p.; regions labelled R.
8. Probability with n: form and solve the quadratic, reject impossible n.
9. Venn: centre outwards; check the total.
10. Pace: 1.5 minutes per mark; leave 10 minutes to check.